Name _____ Period _____ Date _____

6-3 Lesson Quiz: Logarithms

short form

Questions keep their Savvas Lesson Quiz numbers (3, 4, 5), the numbers your packet uses.

3. Which is the logarithmic equation $\log_8 64 = a$, written in exponential form?

Ⓐ $64^a = 8$ Ⓑ $a^8 = 64$ Ⓒ $a^{64} = 8$ Ⓓ $8^a = 64$

4. Select **all** the logarithmic expressions that have been evaluated correctly.

☐ **A.** $\ln 1 = 0$

☐ **B.** $\log_2 9 = 3$

☐ **C.** $\log \frac{1}{100} = \frac{1}{2}$

☐ **D.** $\log_3(-1) = \frac{1}{3}$

☐ **E.** $\log_5 \frac{1}{125} = -3$

Show your thinking. For each expression you did **not** select, write the exponential form that shows it is wrong.

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_____

_____

_____

5. What is the solution to the equation $\log(2x + 4) = 2$? Round to the nearest thousandth, if necessary.

Show your work.

_____

_____

_____

_____

$x =$ _____

Scoring: Question 4 and Question 5 are 10 points each. Question 3 is a bonus of 5 points. Total out of 20.

6-3 Lesson Quiz, short form: answer key

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3. D, $8^a = 64$**.** The base 8 raised to the exponent a gives the argument 64 (and $a = 2$). Distractor A swaps base and argument; B and C put a in the base. Savvas quiz item 3, DOK 2, skill F73. Taught by the form switch in Example 3 (THINK step) and Example 5. **Bonus 5:** D.

4. A and E. A: $e^0 = 1$, so $\ln 1 = 0$. E: $5^{-3} = \frac{1}{125}$, so $\log_5 \frac{1}{125} = -3$. B is wrong: $2^3 = 8$, not 9. C is wrong: $10^{-2} = \frac{1}{100}$, so $\log \frac{1}{100} = -2$. D is wrong: no power of 3 is negative, so $\log_3(-1)$ is undefined. Savvas quiz item 4, DOK 1, skill F69. Example 3 and the packet Practice. **10 points:** 8 for exactly A and E; 4 for one of A or E with no wrong box; 0 otherwise. Plus 2 for at least two correct exponential-form reasons for the unselected choices.

5. $x = 48$**.** Exponential form: $10^2 = 2x + 4$, so $100 = 2x + 4$, $2x = 96$, $x = 48$. Check: $\log(2 \cdot 48 + 4) = \log 100 = 2$. Savvas quiz item 5, DOK 2, skill F69. Example 5 Part B and Try It 5a. **10 points:** 4 for the exponential form $10^2 = 2x + 4$ (or $100 = 2x + 4$); 6 for $x = 48$. A bare 48 with no exponential form earns 6 of 10.

Items left off this form (available for a retake or Form B): Savvas item 1, the inverse of $y = 3^x$ (answer C, $y = \log_3 x$; DOK 2, F69; Example 1, cut, and the packet's top box); Savvas item 2, the best estimate of $\ln 16$ (answer B, 2.77; DOK 2, F73; Example 4, cut; $\ln 16 \approx 2.7726$ by calculator).

IXL replacement: SmartScore 70 or above on 8RK, GBR and BXU replaces this quiz with 20/20.